



GROWTH & GH AXIS

HCG

HCG – 10,000 IU per vial – 10 vials per box

FOR RESEARCH USE ONLY (RUO). NOT FOR HUMAN OR VETERINARY USE.

IDENTIFICATION

CHEMICAL NAME	Human Chorionic Gonadotropin
SYNONYMS / ALIASES	hCG; choriogonadotropin alfa; chorionic gonadotropin
CAS NUMBER	9002-61-3
MOLECULAR FORMULA	Heterodimeric glycoprotein; alpha-subunit approximately 13.4 kDa polypeptide, beta-subunit approximately 22.2 kDa polypeptide; total glycosylated heterodimer approximately 36-40 kDa
MOLECULAR WEIGHT	Approximately 36,700 g/mol (heterodimer, glycosylated)
AMINO ACID SEQUENCE	alpha-subunit (92 residues, identical to LH, FSH, TSH alpha-subunit) non-covalently associated with beta-subunit (145 residues, hCG-specific)
SOURCE	Either purified from pooled human female pregnancy urine (uHCG) or produced recombinantly in mammalian cell culture (rHCG; choriogonadotropin alfa). The specific source for a given batch is identified in batch documentation.

PHYSICAL PROPERTIES

APPEARANCE	Lyophilized white to off-white powder
PURITY SPECIFICATION	Activity-based release (units per vial); chemical purity per batch COA
CONTAINER	10,000 IU per vial, sealed glass vial, rubber-stoppered, aluminum crimp

STORAGE & HANDLING	
UNRECONSTITUTED STORAGE	2-8 C (preferred for glycoprotein stability); some preparations stable at -20 C - confirm against COA
RECONSTITUTED STORAGE	2-8 C, typically stable up to 30 days in bacteriostatic water
SUGGESTED RECONSTITUTION SOLVENT	Bacteriostatic water or sterile water (research use only)
SHELF LIFE (UNRECONSTITUTED, SEALED)	Typically 24 months at 2-8 C
STABILITY NOTES	Glycoprotein heterodimer is sensitive to elevated temperatures and freeze-thaw cycles; the alpha and beta subunits can dissociate under denaturing conditions.

STRUCTURAL NOTES

Human Chorionic Gonadotropin is a heterodimeric glycoprotein hormone composed of two non-covalently associated subunits. The alpha-subunit (92 amino acids) is structurally identical to the alpha-subunit shared by luteinizing hormone (LH), follicle-stimulating hormone (FSH), and thyroid-stimulating hormone (TSH). The beta-subunit (145 amino acids) confers hCG-specific receptor binding via interaction with the luteinizing hormone / chorionic gonadotropin receptor (LHCGR). Both subunits are heavily glycosylated, with N-linked and O-linked oligosaccharide chains contributing substantially to overall molecular mass and circulatory half-life. The activity of preparations is conventionally expressed in International Units (IU) standardized against a reference biological assay.

SELECTED PEER-REVIEWED REFERENCES

1. Cole L.A. - *Biological functions of hCG and hCG-related molecules* - Reproductive Biology and Endocrinology, 2010 - DOI: 10.1186/1477-7827-8-102
2. Stenman U.H. et al. - *The classification, functions and clinical use of different isoforms of HCG* - Human Reproduction Update, 2006 - DOI: 10.1093/humupd/dml029
3. Lapthorn A.J. et al. - *Crystal structure of human chorionic gonadotropin* - Nature, 1994 - DOI: 10.1038/369455a0

References are provided for research background only. Cited publications do not constitute claims about the safety, efficacy, or intended use of this product.

COMPLIANCE DISCLAIMER

This product is sold by Aligned Aminos for laboratory research use only (RUO). It is not a drug, dietary supplement, food, cosmetic, or medical device. It has not been approved by the FDA for any therapeutic, diagnostic, or preventative use. It is not intended for use in humans or animals, and may not be administered, ingested, applied, or otherwise introduced into any living organism. Purchasers represent that they are qualified researchers or laboratory professionals and assume full responsibility for compliance with all applicable laws and institutional review requirements governing the handling of research chemicals.

Batch-specific identity and purity are documented in the Certificate of Analysis (COA) issued for the lot.